# PANJAB UNIVERSITY, CHANDIGARH

Outlines of tests, syllabi and course of reading for Postgraduate Diploma in Disaster Management and Security (Semester System) for the examination of 2010-2011.

| Subject Code | Paper | Marks Th. | Int. Ass. |
|---|---|---|---|
| Semester-I | | | |
| Paper-0I | Disaster Management – Conceptual Framework | 80 | 20 |
| Paper-02 | Risk Factor, Crisis Management & Role of NGOs | 80 | 20 |
| Paper-03 | Disaster Management Structure – Coordination & Response | 80 | 20 |
| Semester-II | | | |
| Paper-04 | Disaster Management – Relief and Rehabilitation | 80 | 20 |
| Paper-05 | War and Disaster Management | 80 | 20 |
| Paper-06 | Project Report + Viva-voce | 70 | +30 |

**For Executive Certificate Course, the candidate will be put to condensed and more focused papers out of the given above which could sharpen their skill in dealing with the subject of Disaster Management.**

**Paper-01: Disaster Management: Conceptual Framework**

M. Marks : 100
Theory : 80 marks
Int. Ass. : 20 marks

**Objective**: - The course will provide general insight to understanding disaster management and approaches including appraising on disaster diplomacy

**Note**: There shall be 9 questions in all. The first question is compulsory containing 15 short answer type questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 10 short answer type questions carrying 20 marks of 2 marks each. Rest of the paper contain 4 units. Each unit shall have two questions and the candidates shall be given a choice of attempting one question from each unit – 4 in all. Each question will carry 15 marks.

## Unit-I

Disaster Management: Concept & significance.

Approaches to disaster management: Rationale, elements and objectives of disaster management.

## Unit-II

Dimensions and typology of Disasters:

Natural disasters – include broad outlines regarding natural and non-natural disasters such as; earthquake, volcanic eruptions, floods, landslides, avalanches, tsunamis, cyclone, climatic change, drought and epidemics

Manmade Disasters – include war, Industrial accidents, soil degradation, desertification, deforestation, depletion of water resources, destruction of ecological system, landslides, fire, oil spill, breakdown of essential services etc.

## Unit-III

National Disaster management policy – significance, principles and policy options.

## Unit-IV

Disaster management in developed and developing nations.
Disaster diplomacy through disaster related factors.

**Books Recommended:**


Bourriau, Janine; **Understanding Catastrophe;** Cambridge University Press, Cambridge, 1992.

Carter, W. Nick; **Disaster Management – A Disaster Manager's Handbook,** A.D.B., Manilla, Philippines, 1991.

Coppola, P. Damon, **Introduction to International Disaster Management,** Elsevier, Oxford, 2006.

Garlake, Teresa, **Dealing with Disasters,** Oxfam, Oxford, 2000.

Goel SL and Ram Kumar, **Disaster Management,** Deep & Deep Publications, New Delhi, 2001.

Goel, S.L. **Encyclopedia of Disaster Management**, Deep and Deep Publications, New Delhi, 2006.

Keller, A Edward, **Environmental Geology,** Prentice Hall, New Delhi, 2010.

Monappa, K.C., **Disaster Preparedness** Akshay Public Agencies, New Delhi, 2004.

Narayan, B., **Disaster Management;** APH Publishing Corporation, New Delhi, 2000.

Sharma V.K., **Disaster Management,** IIPA, New Delhi, 1995.

Singh, Shailendra K, Kundu, Subhash C and Singh, Shobha; **Disaster Management;** Mittal Publications, New Delhi, 1998.

Stephens, Lynn H. and Green, Stephan J; **Disaster Assistant – Appraisal, Reform and New Approaches;** McMillan, New York, 1979.

Turner Barry A and Pidgeon Andnick F, **Man Made Disasters,** Betterworth Heinemann, Oxford, 1997.

**Paper-02: Risk Factor, Crisis Management & Role of NGO**

M. Marks : 100
Theory : 80 marks
Int. Ass. : 20 marks

**Objective**: - The course objective is to explore the risk factor and analyze role of the government and other agencies as tools of crisis management group.

**NOTE:** There shall be 9 questions in all. The first question is compulsory containing 15 short answer type questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 10 short answer type question carrying 20 marks of 2 marks each. Rest of the paper contain 4 units. Each unit shall have two questions and the candidates shall be given a choice of attempting one question from each unit – 4 in all. Each question will carry 15 marks.

## Unit-I

Definition of risk, fundamentals of risk analysis, environmental hazards exposure and risk assessment.

Risk assessment and Disaster mitigation, methodology of stock taking, concept of vulnerability and analysis.

## Unit-II

Identifying potential crisis situations – preparing sound crisis management plan, (crisis communication, stress management, crisis operational guidelines).

## Unit-III

International and Interstate cooperation amongst Non-Government Organizations for Disaster Management.

The role of Corporates, Civil Society/ Community as Non-Government Organizations in handling disaster.

## Unit-IV

Disaster Appreciations based on Mock drill exercises undertaken in collaboration with specialized agencies like Red-Cross and other functionaries.

**Books Recommended:**


Bawa, Noorjahan (ed.) **Non-Government Organisations in Development: Theory and Practice**, Kanishka Publishers, New Delhi, 1997.

Carter, W. Nick, **Disaster Management – A Disaster Manager's Handbook,** A.D.B., Manilla, Philippines, 1991.

Cutter, Susan L. (Ed.): **Environmental Risks and Hazards,** Prentice Hall, New Delhi, 1999.

Garlake, Teresa, **Dealing with Disasters,** Oxfam, Oxford, 2000.

Government of India, **An Action Plan to bring about Collaborative Relationship between Voluntary Organizations and Government**, CAPART, Government of India, New Delhi, 1994.

Government of India/United Nations Development, **Disaster Risk Management Programme (2002-07):** Community Based Disaster Preparedness and Risk Reduction Through Participation of Committees and Local Self Governments.

Kamta Prasad (ed.), **NGOs and Socio-Economic Development Opportunities,** Deep and Deep Publications, New Delhi, 2000.

Meher Nanavaty and Kulkarni P., **NGOs in the Changing Scenario,** Uppal Publishing House, New Delhi,1998.

Modh Satish **Citizens Guide to Disaster Management: How to Save your own life and help others**, McMillan India 2006.

Mutchopadhaya, A.K., **Crisis and disaster management turbulence and aftermath**, Newage International Publications, New Delhi, 2005.

Otto Lerbinger, **The Crisis Manager, Facing risk and responsibility,** Lawrence Erlbaum Associates, New Jersey, 1997.

R.B. Jain, **NGOs in Development Perspective,** Vivek Prakashan, New Delhi, 1995.

Sahni, P. and Ariyabandu, M.M., Malagoda, M. (eds.) **Disaster Risk Reduction in South Asia**, Prentice Hall, New Delhi, 2004.

Singh SS, Pratap Sanjay, National Centre for Disaster Management, **Legislative Framework for Disaster Management (**A study of Legislations in Select Countries), Indian Institute of Public Administration, 2000.

Sinha, P.C. (Ed.): **Encyclopedia of Disaster Management;** Anmol Publications, New Delhi, 1998.

Stephens, Lynn H. and Green, Stephan J; **Disaster Assistant – Appraisal, Reform and New Approaches;** McMillan, New York, 1979.

Wilson, R. and E.A.C. Croouch, **Risk assessment and comparisons: An Introduction**, Science, Washington, 1987.

**Paper-03: Disaster Management Structure – Coordination & Response**

M. Marks : 100
Theory : 80 marks
Int. Ass. : 20 marks

**Objective**: - The course will dwell on the standard operating procedures and contingency planning related to disaster management.

**NOTE:** There shall be 9 questions in all. The first question is compulsory containing 15 short answer type questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 10 short answer type question carrying 20 marks of 2 marks each. Rest of the paper contain 4 units. Each unit shall have two questions and the candidates shall be given a choice of attempting one question from each unit – 4 in all. Each question will carry 15 marks.

## Unit-I

Planning and designing disaster management strategies, policies, schemes, Standard Operating Procedure for enhancing preparedness and capabilities at National and Regional level at the time of disaster/war like situation.

## Unit-II

Allocation and management of resources:

a. Financial aid/ resources.
b. Medical resources.
c. Electrical, mechanical and engineering resources.
d. Communication resources( including its integration with command and control mechanism)
e. Transportation resources.
f. Food and essential commodity resources.
g. Human resources.
h. Infrastructure restoration and build up.

## Unit-III

Disaster Management training – Policies, emergency management teams, managing media and the Role of Information Technology.

Capacity Building and Role of Leadership in Disaster Management.

## Unit-IV

Case Study/examples from recent disasters.

**Books Recommended:**


Delmar, **Fire Fighter's Handbook;** Stanford Thomson Learning (U.S.A.), 2000

Government of India, The Department of Atomic Energy **An Overview of The Emergency Response Plans.(www.dae.gov.in)**

Kirschenbaun, Alan, **Chaos Organization and Disaster Management,** Marcel Dekker, 2004.

Krajeswki & Ritzman, **Operation Management Strategy & Analysis**, Prentice Hall, New Delhi, 2001.

Levitt, Alan M; **Disaster Planning and Recovery,** John Valley and Sons, New York, 1997.

Ministry of Agriculture and Cooperation, Government of India: **National Disaster Response Plan;** New Delhi: 2002.

Ministry of Agriculture and Cooperation, Government of India: **Report of the High Powered Committee on Disaster Management;** New Delhi: 2001.

Misra, Girish K. and Mathur, G.C. (Ed.); **Natural Disaster Reduction,** Reliance Publishing House, New Delhi, 1995.

**National Disaster Response Plan**, NCDM, New Delhi, 2001.

Newburn, Tim; **Disaster and After**, Jessica Kingsley Publishers, London, 1993.

Nuclear Regulatory Commission, **NRC Responses to Events, US,**1998.

P. Michale Vetha Siromony (Ed.), **Source Book on District disaster Management,** Ministry of Agriculture, Government of India, 2000.

Parasuram, S. and Unnikrishnan, P.V. (Ed.), **India Disasters Report – Towards a policy initiative,** Oxford University Press, New Delhi, 2000.

Rao R Ramesh, Jon Eisenberg & Ted Schmitt, **Improving Disaster Management: The Role of IT in mitigation, preparedness, response and recovery**, National Academy of Science, Washington, 2001.

S.N. Charry, **Production & Operation Management,** Tata McGraw Hill, New York, 2005.

Schneid Thomas D. and Collins Larry **Disaster Management and Preparedness,** Lewis Publishers, New York, 2001.

Schnneider, Saundra, K. **Flirting with Disaster: Public Management in Crisis Situations,** M.E. Sharpe,Armonk, New York, 1995.

**UNDRO, Managing Natural Disasters** – A Manual for Policy Makers and Planners, New York, 1991.

## SEMESTER-II

**Paper-04: Disaster Management – Relief and Rehabilitation**

M. Marks : 100
Theory : 80 marks
Int. Ass. : 20 marks

**Objective:** - The paper will study to understand the challenges and issues during post event management/recovery/rehabilitation besides offering strategies used to resolve them.

**NOTE:** There shall be 9 questions in all. The first question is compulsory containing 15 short answer type questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 10 short answer type question carrying 20 marks of 2 marks each. Rest of the paper contain 4 units. Each unit shall have two questions and the candidates shall be given a choice of attempting one question from each unit – 4 in all. Each question will carry 15 marks.

### Unit-I

Disaster Management Policy of India

Post-disaster management measures – Recovery and re-construction measures on medium and long term basis.

## Unit-II

Mobilization of available and non-available resources

Social and economic upliftment measures: capacity building for re-construction and rehabilitation, fostering skills of livelihood, training and awareness programme, medical aid and counselling.

## Unit-III

Relocation and re-construction of housing, buildings, roads, dams, bridges, archives, monuments including water supplies, electricity, communication, waste management.

Socio-economic and psychological effects of disasters on victims.

## Unit-IV

Legal provisions of Disaster Management

**Books Recommended:**


Anderson, James E., **Public Policy Making: An Introduction,** Houghton-Mifflin, Boston, 2005.

Asian Development Bank, **Disaster Mitigation in Asia and the Pacific,** Manila: ADB, 1991.

Collins Larry R. and Scheind Thomas, **Disaster Management and Preparedness.** Taylor and Francis, 2000.

**Disaster Prevention and Mitigation,** UNDRO Publications, Geneva, 1984

Garlake, Teresa, **Dealing with Disasters,** Oxfam, Oxford, 2000.

Green, Stephan, **International Disaster Relief – Towards A Responsive System,** McGraw Hill, New York, 1997.

P. Michale Vetha Siromony (Ed.), **Source Book on District disaster Management,** Ministry of Agriculture, Government of India, 2000.

Schneid Thomas D. and Collins Larry **Disaster Management and Preparedness,** Lewis Publishers, New York, 2001.

Stephens, Lynn H. and Green, Stephan J; **Disaster Assistant – Appraisal, Reform and New Approaches;** McMillan, New York, 1979.

Waugh, William L, **Living with Hazards, Dealing with Disasters,** M.E. Sharpe, New York, 2000.

Waugh, William L, Jr. and Ronald John, **The Functions of Emergency Management, Handbook of Emergency Management**, Greenwood Press, New York, 1990.

**Paper-05: War and Disaster Management**

M. Marks : 100
Theory : 80 marks
Int. Ass. : 20 marks

**Objective:** -The paper will address various aspects of war and non-war situation causing disasters including the role of security forces.

**NOTE:** There shall be 9 questions in all. The first question is compulsory containing 15 short answer type questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 10 short answer type question carrying 20 marks of 2 marks each. Rest of the paper contain 4 units. Each unit shall have two questions and the candidates shall be given a choice of attempting one question from each unit – 4 in all. Each question will carry 15 marks.

## Unit-I

War and Disaster – socio-economic, political, military, health and environmental effects. Lessons learnt and needs identified.

Lethal aspects of conventional wars causing disasters and such like situations.

## Unit-II

Fundamental of nuclear War aspects – includes disasters caused by nuclear weapons, nuclear radiation, nuclear power plant accidents, dirty bombs, nuclear wastes, use of depleted uranium and nuclear terrorism. Protective measures from nuclear disasters.

Fundamentals of Biological war – categorization of biological hazards, communicable diseases and bio-terrorism. Protective measures from biological disasters.

## Unit-III

Fundamentals of Chemical war – dimensions of hazardous chemical material, gases, chemical hazardous waste, chemical terrorism. Protective measures from chemical hazardous.

Terrorism, Weapon of Mass Destructions, suicide attacks, hijacking, landmines, Improvised Explosive Devices, cyber/information warfare, civil strife and internal threats causing disaster like situations.

## Unit-IV

Government preparedness in dealing with disasters during pre-war and post-war period.

Role of Para-military forces and armed forces in dealing with disaster management.

**Books Recommended*:***


Alan M Levitt, **Disaster Planning and Recovery: a guide for facility professionals,** Wiley, New York, 1997.

Bourreair Janine, **Understanding Catastrophe,** Cambridge University Press, Cambridge, 1992.

Cashman, John R; **Hazardous Materials Emergencies – The Professional response Team,** Technomics Publishing Co., U.S.A., 1995.

Kirchenbaum Alan, **Chaos Organisation and Disaster Management**, Marcel Dikker, New York, 2004.

Parasuraman S, Unnikrishnan PV. **India Disaster Report: Towards a policy initiative,** Oxford University Press, New Delhi, 2000.

Pierre Perrin, **Communicable diseases, Handbook of Disaster Medicine,** Ed. Jan de Boer & Marcel, Dubouloz, Hentenaar boek BV, Nieuwegein, Netherlands, 2000.

Stephen Green, **International Disaster Relief – Towards a Responsive System,** McGraw Hill, New York, 1977.

Ursano Robert J, M.D. and Norwood Ann E, M.D. **Trauma and Disaster Responses and Management,** Review of Psychiatry, Volume 22, American Psychiatric Publishing Inc., 2003.

Ursano Robert J., C Fullerton Carol S, Norwood Ann E, **Terrorism and Disaster Individual and Community Mental Health Interventions**, Cambridge University Press, Cambridge, 2003.

Ursano Robert J., Norwood Ann E, Fullerton Carol S, **Bioterrorism Psychological and Public Health Interventions**, Cambridge University Press, Cambridge, June, 2004.

Wahlstron Bjorn, **Radiation Health and Society,** Louise Fichtinger: International Atomic Energy Agency, 1998.

**Paper-06: Project Report and Viva-Voce**

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|---|---|
| M. Marks | : 100 |
| Project Report | : 70 Marks |
| Viva Voce | : 30 Marks |

**Objective**: - The paper will be compulsory and shall include sand model discussion, scenario building, black board exercises, management games and exercises on various types of disasters of both natural and manmade categories. A Project Report will be prepared. It will be based upon presentations, discussions and demonstrations during their visits to various headquarters, Departments/Units of government and non-government agencies.

A viva examination will be conducted based on the project report.